

Research

The need for fairness and transparency in organizations has drawn a lot of attention to salary prediction and compensation management in recent years. Even when employees have comparable qualifications, research indicates that manual salary decisions may result in bias based on gender, background, or subjective evaluation. Organizations are using data-driven and machine learning-based models to support objective salary determination in order to overcome this.

Prior research has shown that a number of factors, including prior performance, education level, work experience, and skill proficiency, are important in determining employee compensation. Predictive models can be trained to discover recurring salary patterns by utilizing past employee and applicant data. These methods ensure adherence to organizational and ethical standards, encourage equal compensation for comparable profiles, and lessen human intervention.

Analysis

The analysis focuses on looking at past data from Company X to find connections between salary outcomes and employee characteristics. To increase model accuracy and dependability, data preprocessing methods like handling missing values, normalization, and feature selection are used. Understanding which characteristics have the greatest influence on salary decisions is made easier with correlation analysis.

Regression-based machine learning models are used to forecast salary offers for new applicants based on the data analysis. The model guarantees that salary recommendations are comparable for candidates with comparable experience and skill sets. This analytical approach minimizes discrimination, enhances consistency in salary decisions, and supports HR teams with a fair and automated salary recommendation system.

Ideate

The proposed ideation focuses on developing a simple, beginner-friendly salary prediction system that supports the HR department of Company X in making fair and consistent salary decisions. The core idea is to reduce human bias by replacing

subjective judgment with a data-driven model that learns salary patterns from historical employee data. By automating salary recommendations, the system ensures that employees with similar profiles—such as experience, education, job level, and department—are offered comparable compensation.

To demonstrate the concept, synthetic employee data is generated to resemble a real HR dataset. This allows the model to be trained and tested even in the absence of sensitive real-world salary records. Key employee attributes are selected as predictors, while sensitive attributes like gender are deliberately excluded from the model training to prevent discriminatory learning. This design choice aligns with the project goal of eliminating bias and promoting fairness in salary allocation.

The ideation employs a machine learning regression approach using a Random Forest model, chosen for its ability to capture non-linear relationships between employee attributes and salary. The model is trained on historical patterns and evaluated using error metrics to ensure reliable salary predictions. A basic fairness check is also incorporated by comparing predicted salaries across genders, reinforcing ethical salary practices.

Finally, the system includes an interactive feature that allows HR personnel to input a new candidate's profile and instantly receive a recommended salary offer. This transforms the model into a practical decision-support tool that can be easily adopted in real recruitment scenarios. Overall, this ideation demonstrates how a simple and transparent machine learning solution can help Company X achieve equitable, consistent, and unbiased salary determination.

Build

Algorithm

Step 1: Start

Initialize required Python libraries for data handling, machine learning, and evaluation.

Step 2: Data Collection

Collect a synthetic employee dataset consisting of:

- IDX
- - Applicant_ID
- - Total_Experience
- - Total_Experience_in_field_applied
- - Department
- - Role
- - Industry
- - Organization
- - Designation
- - Education
- - Graduation_Specialization
- - University_Grad
- - Passing_Year_Of_Graduation
- - PG_Specialization
- - University_PG
- - Passing_Year_Of_PG
- - PHD_Specialization
- - University_PHD
- - Passing_Year_Of_PHD
- - Curent_Location
- - Preferred_location
- - Current_CTC
- - Inhand_Offer
- - Last_Appraisal_Rating
- - No_Of_Companies_worked
- - Number_of_Publications
- - Certifications
- - International_degree_any

Ensure salary values lie within a realistic range to avoid outliers.

Step 3: Data Preparation

Store the generated employee records in a structured data frame for further processing and analysis.

Step 4: Categorical Data Encoding

Convert categorical attributes (education, job level, department) into numerical values using label encoding so that the machine learning model can process them.

Step 5: Feature Selection

Select relevant features influencing salary:

- Experience
- Encoded education
- Encoded job level
- Encoded department

Separate the dataset into:

- Input features (X)
- Target variable (salary)

Step 6: Data Splitting

Split the dataset into training and testing sets using an 80:20 ratio to evaluate model performance on unseen data.

Step 7: Model Training

Train a Random Forest Regression model using the training dataset to learn salary patterns from historical employee data.

Step 8: Model Evaluation

Predict salaries on the test dataset and compute the Mean Absolute Error (MAE) to measure prediction accuracy.

Step 9: Fairness Evaluation

Analyze predicted salaries across gender groups by:

- Calculating the average predicted salary for male and female employees
- Measuring the salary gap
- Flagging potential bias if the difference exceeds a predefined threshold

Step 10: New Candidate Salary Prediction

Accept new candidate details as user input:

Convert inputs into numerical form and predict the recommended salary using the trained model.

Step 11: Display Output

Display the predicted salary offer along with the candidate's profile details for HR decision support.

Step 12: End

Conclude the process with a salary recommendation ready for approval.

Code

```
# Expected CTC Project

# Shreeya Jagtap


# Step 1: Mount Google Drive

from google.colab import drive

drive.mount('/content/drive')
```

```
# Step 2: Import libraries pandas and os

import pandas as pd

import os


# Step 3: Load dataset from Google Drive

folder_path = "/content/drive/My Drive/ColabData"


file_name = None

for f in os.listdir(folder_path):

    if f.endswith(".csv") or f.endswith(".xlsx"):

        file_name = f

        break


if file_name is None:

    raise Exception("No Excel or CSV file found")


file_path = folder_path + "/" + file_name


if file_name.endswith(".csv"):

    data = pd.read_csv(file_path)

else:

    data = pd.read_excel(file_path)
```

```
print("\nDataset Loaded Successfully!")

print(data.head())


# Step 4: Handle missing values

data = data.fillna(data.mode().iloc[0])


# Step 5: Identify target column (Expected CTC)

target_column = data.columns[-1] # last column omitted

feature_columns = data.columns[:-1]


print("\nInput Parameters Required:")

for col in feature_columns:

    print("-", col)


# Step 6: Taking user input for all columns

user_input = {}


print("\nEnter Candidate Details:")

for col in feature_columns:

    value = input(f"Enter {col}: ")

    user_input[col] = value
```

```

# Convert user input to DataFrame

user_df = pd.DataFrame([user_input])


# Step 7: Convert data types to match dataset

for col in feature_columns:

    try:

        user_df[col] = user_df[col].astype(data[col].dtype)

    except:

        pass


# Step 8: Find similar profiles (exact match)

filtered_data = data.copy()


for col in feature_columns:

    filtered_data = filtered_data[filtered_data[col] ==
user_df[col].iloc[0]]


# Step 9: If no exact match, relax filtering (ignore one column at a time)

if filtered_data.empty:

    filtered_data = data.copy()

    for col in feature_columns[:-1]:

        filtered_data = filtered_data[filtered_data[col] ==
user_df[col].iloc[0]]

        if not filtered_data.empty:

```



```

        break

# Step 10: Calculate Expected CTC

if filtered_data.empty:

    print("\nNo similar profile found.")

    print("Using overall average Expected CTC.")

    expected_ctc = data[target_column].mean()

else:

    expected_ctc = filtered_data[target_column].mean()


# Step 11: Display result

print("\n-----")

print("Predicted Expected CTC:", int(expected_ctc))

print("Based on", len(filtered_data), "similar employee profiles")

print("-----")

```

Implementation

```
Enter Candidate Details:
Enter IDX: 1
Enter Applicant_ID: 12068
Enter Total_Experience: 4
Enter Total_Experience_in_field_applied: 2
Enter Department: HR
Enter Role: Consultant
Enter Industry: Training
Enter Organization: g
Enter Designation: HR
Enter Education: PG
Enter Graduation_Specialization: Arts
Enter University_Grad: Pune
Enter Passing_Year_Of_Graduation: 2020
Enter PG_Specialization: others
Enter University_PG: Pune
Enter Passing_Year_Of_PG: 2013
Enter PHD_Specialization: chemistry
Enter University_PHD: 2022
Enter Passing_Year_Of_PHD: 2026
Enter Current_Location: pune
Enter Preferred_location: pune
Enter Current CTC: 3500
Enter Inhand_Offer: n
Enter Last_Appraisal_Rating: 5
Enter No_Of_Companies_worked: 2
Enter Number_of_Publications: 2
Enter Certifications: 5
Enter International_degree_any: 0
```

```
-----
Predicted Expected CTC: 384551
Based on 1 similar employee profiles
-----
```

Conclusion

This project successfully demonstrates a data-driven approach to determining the expected CTC for prospective candidates while minimizing human bias in salary decisions. By utilizing historical employee data stored in the Excel dataset, the system collects candidate details across all relevant parameters and identifies similar employee profiles through grouping and filtering techniques. The expected CTC is then calculated based on the average salary offered to candidates with comparable profiles, ensuring consistency and fairness in compensation.

The proposed solution reduces the reliance on manual judgment and the risk of discrimination by relying on objective data rather than subjective decision-making. It also provides transparency in the salary determination process, as the final offer can be clearly justified using records. Overall, this project presents a straightforward, robust, and easily comprehensible mechanism suitable for real-world HR applications, particularly in organizations seeking to maintain equitable salary structures for employees with similar qualifications, experience, and skill sets.

Referance

[https://www.ijert.org/predictive-analysis-of-hr-salary-using-machine-learning-techniques?](https://www.ijert.org/predictive-analysis-of-hr-salary-using-machine-learning-techniques?utm_source=chatgpt.com)

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